

Department of Mathematics, University of Maryland

Probability graduate written examination

August 2025

Instructions to the student:

- Answer all six problems. Each will be graded from 0 to 10.

In each problem, if you cannot prove one of the statements, it is still fine to use it for the purpose of proving the other statement(s). You will get partial credit in this case. Different parts of the same problem may or may not require the use of the result(s) claimed in the earlier part(s).

- Use a different booklet for each problem. Write the problem number and your code number (NOT YOUR NAME) on the outside cover.
- Keep scratch work on separate pages in the same booklet.
- If you use a “well known” theorem in your solution to any problem, it is your responsibility to make clear which theorem you are using and to justify its use.

1. Let $(X_n)_{n=1}^{\infty}$ and X be random variables.

- (a) Prove that if (X_n) converges to X almost surely then (X_n) converges to X in probability.
- (b) Prove that if (X_n) converges to X in probability then (X_n) converges to X in distribution.

Remark: While these are well known facts, you need to supply their proof and cannot just rely on the fact that they are known.

2. Calculate (justifying all steps):

(a)

$$\lim_{n \rightarrow \infty} \int_{-\frac{1}{2}}^{\frac{1}{2}} \int_{-\frac{1}{2}}^{\frac{1}{2}} \cdots \int_{-\frac{1}{2}}^{\frac{1}{2}} \cos \left(\frac{x_1 + \cdots + x_n}{n} \right) dx_n dx_{n-1} \cdots dx_1. \quad (1)$$

(b)

$$\lim_{n \rightarrow \infty} \int_{-\frac{1}{2}}^{\frac{1}{2}} \int_{-\frac{1}{2}}^{\frac{1}{2}} \cdots \int_{-\frac{1}{2}}^{\frac{1}{2}} \cos \left(\frac{x_1 + \cdots + x_n}{\sqrt{n}} \right) dx_n dx_{n-1} \cdots dx_1. \quad (2)$$

3. Let $(X_n)_{n=1}^\infty$ be independent and identically distributed random variables, with distribution

$$\mathbb{P}(X_n = -2) = \mathbb{P}(X_n = 2) = 1/2. \quad (3)$$

Denote, for each integer $m \geq 1$,

$$A_m := \frac{X_1 + \cdots + X_m}{m}. \quad (4)$$

- (a) Prove that the limit $\lim_{n \rightarrow \infty} \mathbb{E}((A_n)^n)$ does not exist.

Hint: Consider even and odd n separately.

- (b) Prove that, for each integer $m \geq 1$,

$$\mathbb{P}(|A_m| \geq 1/2) \leq \frac{16}{m}. \quad (5)$$

- (c) Prove that

$$\lim_{n \rightarrow \infty} \mathbb{E}((A_{3^n})^n) = 0. \quad (6)$$

4. Let $(s_n)_{n=1}^\infty$ be independent and identically distributed, with

$$\mathbb{P}(s_n = 1) = \mathbb{P}(s_n = -1) = 1/2. \quad (7)$$

Define a process $(X_n)_{n=0}^\infty$ by $X_0 = 1/2$ and, for each $n \geq 0$,

$$X_{n+1} = X_n + \frac{1}{2}s_{n+1}(1 - X_n)(1 + X_n). \quad (8)$$

- (a) Prove that the limit $\lim_{n \rightarrow \infty} X_n$ exists almost surely and in L_1 .

Hint: Observe (and prove) that $|X_n| \leq 1$ for all n . Identify what type of process X is.

- (b) Denote the limiting random variable from the previous item by X_∞ . Find the distribution of X_∞ (and justify your claims).

5. Stella King is playing “Candy Birds”, a game with four levels (levels 1, 2, 3, 4). Stella starts in level $X_0 = 2$. Stella wins each round with probability $2/3$ and loses with probability $1/3$, independently between rounds. When winning, she progresses one level higher, unless she is already at level 4, in which case she stays at level 4. When losing, she drops one level lower, unless she is already at level 1, in which case she stays at level 1. We let X_n record Stella’s level after playing n rounds.

For example, if Stella wins a round, then loses 3 consecutive rounds, then wins another round, then $X_0 = 2, X_1 = 3, X_2 = 2, X_3 = 1, X_4 = 1, X_5 = 2$.

- (a) Let $T := \min\{n : X_n = 4\}$ be the first round that Stella reaches level 4. Calculate $\mathbb{E}T$ (and justify your arguments).

- (b) Calculate the average level that Stella plays in the long run. That is, calculate the limit $\lim_{n \rightarrow \infty} \mathbb{E}(X_n)$ (and justify your arguments).

Remark: Once a numerical expression for the answers is reached, it is not required that you reduce it to a “neat form” as a rational or decimal fraction. An expression consisting of a sum of a few explicit terms is fine as a final answer.

6. Let $(B_t)_{t \geq 0}$ be a standard Brownian motion. For each $a \geq 0$ denote

$$T_a := \min\{t \geq 0: B_t = a\}. \quad (9)$$

It is known that T_a is almost surely finite.

- (a) For each $a, s > 0$, prove that

$$T_{sa} \stackrel{d}{=} s^2 T_a. \quad (10)$$

- (b) For each $a, b > 0$, prove that

$$T_{a+b} \stackrel{d}{=} T_a + T'_b \quad (11)$$

where T'_b has the distribution of T_b and is independent of T_a (and where $\stackrel{d}{=}$ denotes equality in distribution).